

Six-section DLPFC spatial panel: layer and partition profiles (1 of 3)

Donor-macro profiles after section-within-donor aggregation; mean $\pm$ sample SD across five algorithmic seeds where applicable.

Representation	ARI	NMI	Cluster SIL	Reference SIL	Purity
Harmony	0.1635	0.2300	0.0550	-0.0177	0.4819
Harmony + GenoRefine	0.1711 $\pm$ 0.0233	0.2119 $\pm$ 0.0117	0.0427 $\pm$ 0.0068	-0.0284 $\pm$ 0.0012	0.4469 $\pm$ 0.0038
SpaGCN	0.1457 $\pm$ 0.0157	0.3103 $\pm$ 0.0237	0.2774 $\pm$ 0.0080	0.0060 $\pm$ 0.0137	0.5650 $\pm$ 0.0229
GraphST	0.4206 $\pm$ 0.0086	0.5677 $\pm$ 0.0057	0.1289 $\pm$ 0.0011	0.0871 $\pm$ 0.0004	0.7668 $\pm$ 0.0007

Harmony is a single fixed embedding after 10 outer iterations. ARI/NMI use a shared fixed Leiden resolution (0.5), but realized cluster counts differ; interpret them as fixed-resolution profiles.

Six-section DLPFC spatial panel: local spatial structure and mixing (2 of 3)

Donor-macro profiles after section-within-donor aggregation; mean $\pm$ sample SD across five algorithmic seeds where applicable.

Representation	Spatial J	Physical px	Physical/local	Clusters	iLISI	D_{batch}
Harmony	0.0125	2436.2513	17.2711	5.28	0.8626	1.9323
Harmony + GenoRefine	0.0102 $\pm$ 0.0004	2595.7261 $\pm$ 22.3291	18.4093 $\pm$ 0.1589	3.20 $\pm$ 0.45	0.8781 $\pm$ 0.0027	1.9477 $\pm$ 0.0025
SpaGCN	0.0144 $\pm$ 0.0027	2784.7047 $\pm$ 99.4488	19.7433 $\pm$ 0.7122	13.26 $\pm$ 1.21	NA	NA
GraphST	0.0723 $\pm$ 0.0007	1509.3995 $\pm$ 3.2148	10.7452 $\pm$ 0.0220	6.50 $\pm$ 0.20	0.8356 $\pm$ 0.0008	1.9877 $\pm$ 0.0003

Lower physical/local ratio is favorable. SpaGCN mixing is NA because it is fit per section. GraphST uses donor-pair PASTE alignment; common tasks are therefore intentionally compared with explicit asymmetry.

Six-section DLPFC spatial panel: secondary native partitions (3 of 3)

Task-native endpoints are kept separate from the common fixed-resolution evaluator.

Method	Native partition	ARI	NMI	Clusters
SpaGCN	Predicted	0.2222 ± 0.0105	0.3471 ± 0.0187	NA
SpaGCN	Hex-refined	0.2465 ± 0.0106	0.3785 ± 0.0166	NA
GraphST	mclust	0.4590 ± 0.0255	0.6197 ± 0.0175	6.33 ± 0.00
GraphST	Spatial-refined	0.4762 ± 0.0261	0.6564 ± 0.0146	6.33 ± 0.00

Values are mean $\pm$ sample SD across five descriptive algorithmic seeds after section-to-donor-to-macro aggregation. SpaGCN cluster counts are fixed by its section-specific label-free K and are omitted here; GraphST donor-pair K values are 6, 7, and 6.